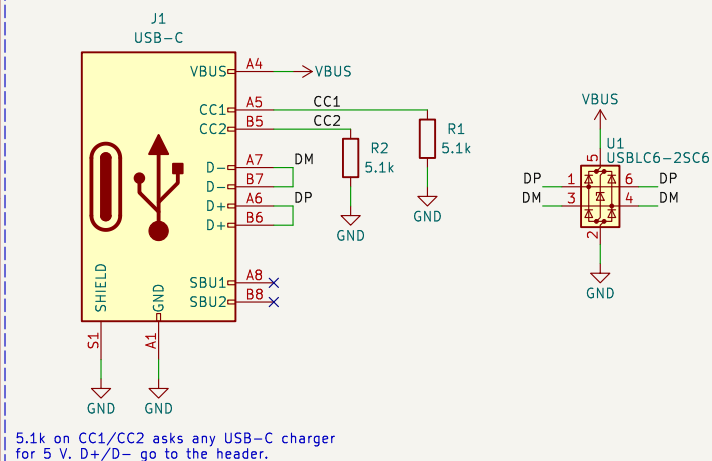


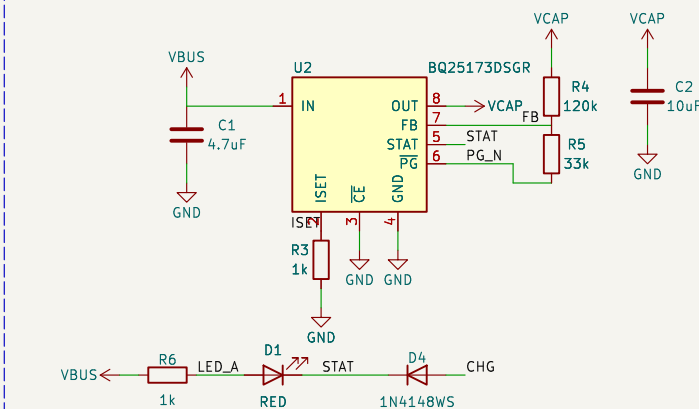
USB-C input



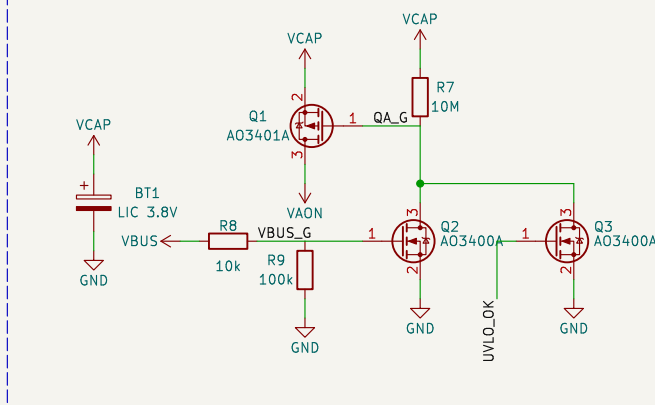
5.1k on CC1/CC2 asks any USB-C charger for 5 V. D+/D- go to the header.

Charger – TI BQ25173

| 300 mA constant current, then constant voltage.
| VREG = 0.8 V x (1 + R4/R5): LIC 3.71 V (R4 120k) / EDLC 2.62 V (R4 75k).
| R5 returns to /PG: no divider drain while USB is unplugged.

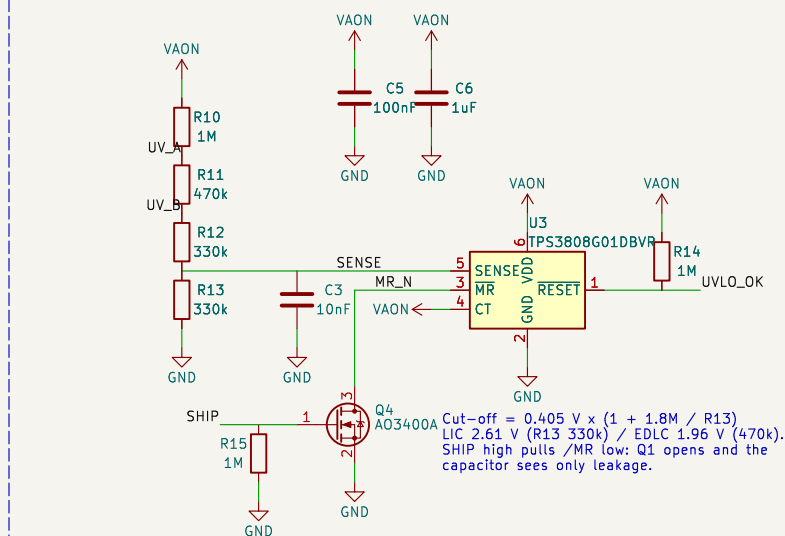


Capacitor + master switch



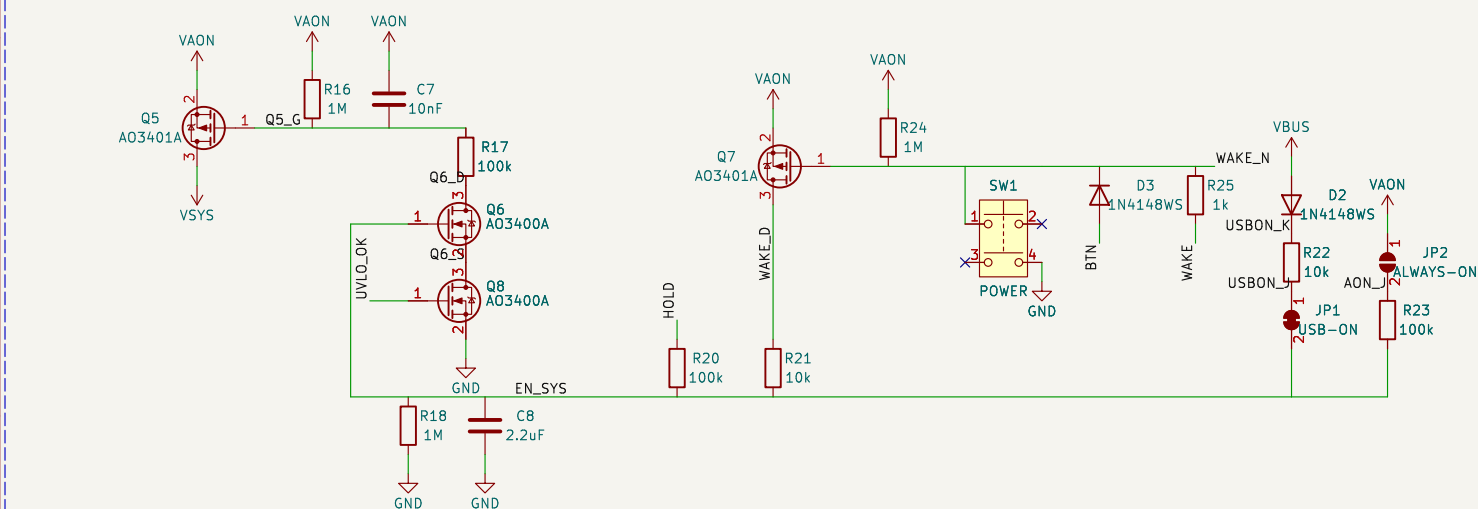
Q1 connects the capacitor to everything else.
USB (Q2) turns it on; the supervisor (Q3) keeps
it on. Below the cut-off Q1 opens and stays open
until USB returns.

Cut-off + ship mode – TPS3808



Cut-off = $0.405 \text{ V} \times (1 + 1.8\text{M} / \text{R13})$
 LIC 2.61 V (R13 330k) / EDLC 1.96 V (470k).
 SHIP high pulls /MR low: Q1 opens and the capacitor sees only leakage.

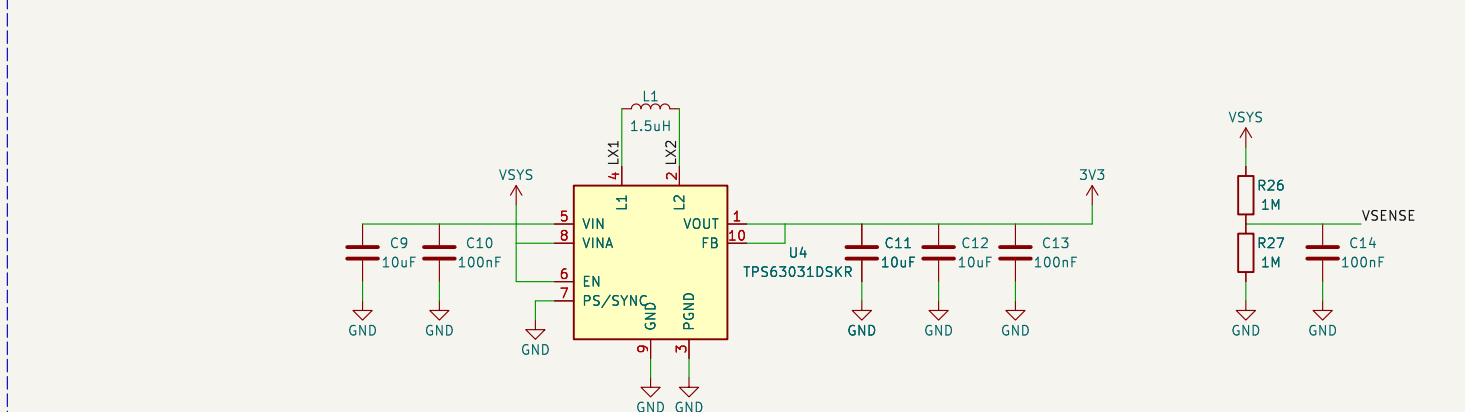
Power switch + wake logic



Q5 feeds VSYS while EN_SYS is high. EN_SYS is raised by the button, the RTC alarm, USB (JP1, default on), HOLD from firmware, or JP2 (always on). C8 holds it ~2 s after a press so firmware can raise HOLD; HOLD low = off in 0.2 s.

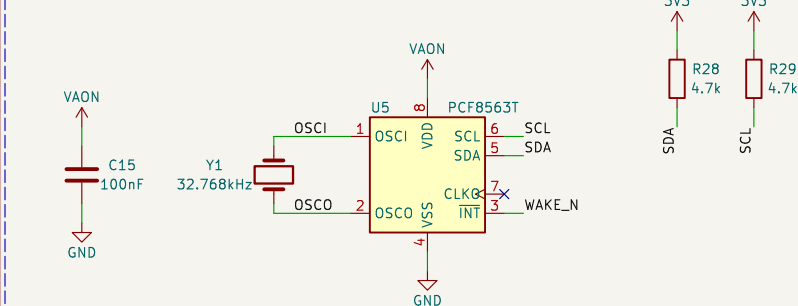
Q8 (gated by the supervisor) sits under Q6: below the cut-off nothing turns it on.

3.3 V buck-boost – TI TPS63031



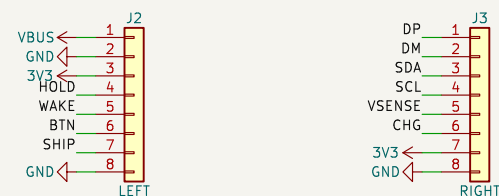
1.8–5.5 V in, 3.3 V out, ~500 mA from a 2.6 V capacitor.
VSENSE = VSYS / 2 for your ADC; draws current only while on.

Real-time clock – NXP PCF8563



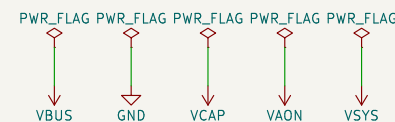
Runs from VAON. Its alarm pulls WAKE_N low to start the gadget. I2C address 0x51.

Headers (0.1 in, 0.7 in apart)



Pin 1 of each header is at the USB-C end.

Power flags (ERC only)



CapCore v1.0 – supercapacitor power module

Build variants differ only in R4 and R13:
 LIC (3.8 V hybrid supercap): R4 120k, R13 330k
 charge 3.71 V, cut-off 2.61 V
 EDLC (2.7 V supercap): R4 75k, R13 470k
 charge 2.62 V, cut-off 1.96 V
 Never fit a 2.7 V EDLC to a board built
 with the LIC values.

Standby (gadget off) ~5 uA; cut-off and
firmware: raise HOLD at boot, drop it to
power off.

Generated by scripts/gen_schematic.py from scripts/netlist.py
Supercapacitor power module: USB-C charging, cut-off, 3.3 V

CapCore
Sheet: /
File: capcore.kicad_sch

Title: CapCore – supercapacitor power module

Size: A3	Date: 2026-09-25
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KiCad E.D.A.	kiCad-cli 7.0.11+dfsg-1build4
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Rev: 1.0

Id: 1/1